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Agricultural Health and Safety: Evaluating Farmers' Knowledge, Attitude, and Safety Behavior in Northern Thailand

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ABSTRACT

Background: Agriculture is vital for economic growth and livelihoods in northern Thailand, yet it is challenged by hazardous practices such as improper chemical use and machinery accidents. This study aimed to evaluate the risk and determine the relationship between pesticide exposure and knowledge, attitudes, and safety behaviors.

Methods: This cross-sectional study was conducted between October and December 2017 in the San Pa Tong district of Chiang Mai, Thailand. Data were collected through structured questionnaires and face-to-face interviews with 280 farmers.

Results: The findings indicated that knowledge and safety behavior scores varied significantly with educational attainment and training received. Farmers with higher education levels or those who had undergone training exhibited better knowledge and safer behavioral practices, whereas income levels did not significantly affect knowledge or safety behaviors. The pesticide exposure risk level was associated with the study variables, with those recognizing a 'normal' risk level demonstrating greater knowledge and safer behaviors than those who reported the risk level as safe or at risk. A strong positive correlation between knowledge and safety behavior was observed, indicating that increased awareness can result in more cautious practices. These findings suggest that educational and training programs are crucial for promoting safer pesticide use among farmers. Although attitudes toward pesticide safety varied based on income, education, and training positively influenced knowledge and behavior.

Conclusion: The study concludes that understanding farmers' perceptions of pesticide risks is essential for designing and implementing effective educational interventions.

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1. Introduction

Agriculture is a vital sector in Thailand [1], particularly in the northern regions, contributing significantly to the country's economy and providing livelihoods for millions of people. Farming is the main activity in these regions, and the majority of residents are engaged in this occupation. Farming in Northern Thailand primarily revolves around crops such as rice, corn, onion, and fruits, as well as livestock [2]. Despite the importance of agriculture, several challenges plague this sector, including safety-related issues. Safety behavior in agriculture refers to the actions taken by farmers to protect themselves and their environment from potential hazards such as the misuse of agricultural chemicals, dangerous farming practices, and machinery accidents [3].

A systematic review of agricultural pesticide exposure in Thailand from 2009 to 2016 found that agricultural workers had

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many health issues, mostly from exposure to organophosphates [4]. A study on pesticide use among Thai farmers found widespread exposure to organophosphates and carbamates in northern Thailand, primarily through dialkylphosphate metabolites. Biomarkers such as acetylcholinesterase and butyrylcholinesterase, used to detect exposure, showed lower enzyme levels in farmers than in non-farmers [5]. Similarly, the Bureau of Occupational and Environmental Diseases, Ministry of Public Health, utilizes plasma acetylcholinesterase for occupational health services in primary care units and community hospitals across the country to evaluate the health risks that affect agricultural workers. Using fingertip blood samples, workers were interviewed and tested to determine plasma acetylcholinesterase levels inhibited by organophosphate and carbamate pesticides [6]. The National Health Security Office indicates that between 2016 and 2017, poisoning cases from agricultural hazardous substances increased, averaging 5,005 patients and 597 deaths annually [7].

Existing research has shown that farmers' knowledge and attitudes play a crucial role in determining their safety behaviors [8]. In other contexts, better knowledge of best practices, awareness of risks, and a positive attitude toward safety have been found to enhance safety behavior [9]. However, there is a relative lack of data and research focused specifically on Northern Thailand, which has a unique context owing to its specific farming practices, local conditions, and cultural background. Understanding farmers' knowledge and attitudes toward safety practices is crucial in promoting safer farming practices and ultimately enhancing the well-being of farmers and their communities [10,11]. It is critical to gain insights into the factors that influence these behaviors, which can help design interventions and policies that are more effective in promoting safe practices [12].

Thailand's agricultural sector faces significant safety challenges, including the improper use of pesticides and fertilizers, inadequate use of safety equipment, and unsafe machinery practices [6]. These issues underscore the substantial gap in safety knowledge and practices among farmers in Northern Thailand. This study aims to address this gap by exploring the factors influencing the safety behaviors of these farmers. It involves assessing the risk of pesticide exposure through in-field tests and analyzing the impact of demographic variables, education, income, and training on farmers' safety knowledge, attitudes, and practices. These findings could help to enhance training programs, shape government policies, and implement interventions to promote safer farming and improve public health in the region.

2. Materials and methods

2.1. Study participants and research materials

This study adopted cross-sectional research conducted between October and December 2017 in the Sanpatong district of Chiang Mai, Thailand. There were 9,323 agricultural families registered in the Sanpatong district agricultural extension office; unfortunately, the number of farmers has not been recorded [13]. The G*Power 3.1 software was utilized to calculate the number of participants. To achieve a study power of 95% with a type I error rate of 0.05 and an effect size of 0.23, a sample size of 266 individuals is required. After adjusting for a 5% attrition rate, the final sample size was 280 participants. Recruitment utilized collaboration with local agricultural officers and community leaders, employing purposive sampling to identify active farmers aged 18 and above. Not all farmers were reachable, but those fulfilling inclusion criteria and expressing willingness to participate were incorporated, with noted non-participation reasons including time limitations and health concerns. The participants should have had no disease that affected the level of cholinesterase activity, such as liver disease, anemia, glaucoma, malnutrition, etc.

2.2. Data collection

2.2.1. Questionnaires survey

Trained research assistants used structured questionnaires during face-to-face interviews. The questionnaires were systematically organized into four key sections to gather a comprehensive range of data.

Demographics: This foundational section aimed to collect essential background information about participants, including age, sex, educational level, and farming experience. Information about the type of farming activities in which participants were involved was also collected, providing context for their responses in the subsequent sections.

Knowledge was evaluated by examining participants' comprehension of secure agricultural practices, encompassing farming hazards, appropriate chemical application techniques, and the utilization of personal protective equipment. This was conducted through multiple-choice and true/false inquiries, with correct responses yielding scores ranging from 0.5 to 1 point, thereby allowing for a maximum attainable score of 33 points.

Attitude was assessed through a Likert scale methodology, whereby the response "Agree" garnered 2 points for indicating safe attitudes, "Disagree" received 2 points for reflecting unsafe attitudes (where applicable), and "Not sure" accrued 1 point, culminating in a total potential score of 40 points.

Safety behavior was scrutinized based on the prevalence and comprehensiveness of secure practices, such as personal protective equipment utilization and adherence to regulatory standards, with questions being appraised on a scale of 0 to 2 points, resulting in an aggregate score of 56 points.

Each score was derived by aggregating the individual responses to systematically quantify participants' knowledge, attitudes, and behaviors on agricultural safety.

The questionnaire was modified from our previous study which originated from the farmers' insecticide exposure risk assessment questionnaire by the Ministry of Public Health, Department of Disease Control, Bureau of Occupational and Environmental Diseases [14]. The pre-test was performed with 30 participants similar to the sample group, yielding a Cronbach alpha of 0.96 to ensure validity and reliability. Furthermore, the questionnaire was translated into the northern local language by local interviewer staff during the interviews to eliminate language barriers and facilitate clear understanding among participants.

2.2.2. Assessment of blood cholinesterase levels

Cholinesterase test kits (reactive paper) purchased from the Government Pharmaceutical Organization were used to measure enzyme activity and assess the level of risk associated with pesticide exposure, specifically organophosphates and carbamates. Blood samples were obtained via finger puncture with the supervision of a trained person during the blood sample collection with the supervision of a trained person before questionnaires were collected. The outcome was determined based on the color of the reactive paper seven minutes after the addition of the plasma sample. The observed colors, yellow, yellow–green, green, and dark green, correspond to normal, safe, risky, and unsafe conditions, respectively.

2.3. Data analysis

Data were analyzed using the Statistical Package for the Social Sciences, version 17. The demographic characteristics were

Table 1
Distribution of demographic characteristics of the study population

Demographical characteristics		Frequency ($n = 280$)	Percentage (%)
Sex	Man	105	37.5
	Woman	175	62.5
Age (years)	≤41	19	6.8
	41–50	71	25.4
	51–60	119	42.5
	≥61	71	25.4
Education levels	No education	14	5.0
	Primary school	183	65.4
	Junior high school	48	17.1
	Senior high school	22	7.9
	Higher than high school	13	4.6
Income (baht/month)	≤4000	116	41.4
	4,001–8,000	95	33.9
	≥8,001	69	24.6
Duration in agricultural (years)	≤10	100	35.7
	11–20	114	40.7
	≥21	66	23.6

summarized using descriptive statistics. Because the data were not normally distributed, to examine the associations between knowledge, attitude, and safety behavior scores with demographic characteristics and the level of pesticide risk, the study utilized the Mann–Whitney U test and multivariate regression analysis. The association between farmers' knowledge of and attitudes toward safety behavior was assessed using partial correlation analysis and Spearman's rho.

2.4. Ethical considerations

The study was approved by an ethical review board prior to data collection (Doc No. 1/2560) at the Research Institute for Health Sciences, Chiang Mai University. Consent was obtained from all participants after they were informed of the study's purpose.

3. Results

3.1. Sociodemographic and characteristics

As shown in Table 1, 175 participants (62.5%) were female and 105 (37.5%) were male. The age of the participants ranged from less than or equal to 41 to over 61 years. The majority of participants (42.5%) were within the age range of 51–60 years, followed by those within the age range of 41–50 years and those over 61 years, both representing 25.4% of the total participants. A small proportion of participants (6.8%) were aged 41 years or younger. In terms of education, most participants (65.4%) had received primary school education. This was followed by junior high school (17.1%), senior high school (7.9%), no education (5.0%), and higher than high school (4.6%). Income levels varied among the participants. A substantial proportion (41.4%) reported an income of 4,000 baht or less per month. Nearly 34% (33.9%) of the participants earned between 4,001 and 8,000 baht per month and 24.6% earned over 8,000 baht per month. In terms of duration in agriculture, the majority of participants (40.7%) had been involved in agriculture for 11–20 years, followed by those aged 10 years or less (35.7%), and those aged 21 years or more (23.6%).

This demographic analysis provides an overview of the surveyed population, which is primarily composed of middle-aged to older farmers, primarily educated till the primary school level, with females being the most distributed between the two genders. Most participants were in the lower-to-middle-income brackets.

The multivariate regression analysis revealed several significant findings regarding the predictors of knowledge, attitude, and safety behavior scores among farmers as shown in Table 2. For knowledge

Table 2
Multivariate regression analysis results for knowledge, attitude, and safety behavior scores by demographic characteristics among farmers

Demographical characteristics		Median (IQR)		
		Knowledge	Attitude	Safety behavior
Sex	Man	18(12.0–25.5)	30(26.0–32.0)	45(23.0–50.0)
	Woman	16(8.0–24.0)	30(27.0–32.0)	38(23.0–49.0)
	B (SE)	-3.142 (1.711)	0.837 (0.611)	-2.069 (1.643)
	p	0.067	0.172	0.209
Age (years)	≤ 41	13(7.0–21.0)	30(27.0–32.0)	24(21.0–47.0)
	41–50	20(11.0–25.0)	30(26.0–33.0)	42(23.0–50.0)
	51–60	16(8.0–25.0)	30(26.0–32.0)	42(23.0–49.0)
	≥61	17(11.0–25.0)	30(28.0–32.0)	45(24.0–50.0)
	B (SE)	1.146 (0.984)	0.370 (0.351)	1.944 (0.945)
	p	0.245	0.293	0.041*
Education levels	No education	14(11.0–23.3)	28(24.0–34.5)	45(25.0–50.3)
	Primary school	16(9.0–25.0)	30(26.0–32.0)	43(23.0–49.0)
	Junior high school	18(8.5–24.0)	30(29.0–32.0)	33(21.5–49.0)
	Senior high school	20(8.0–26.5)	32(29.8–34.3)	28(21.0–50.0)
	Higher than high school	26(20.0–30.0)	33(30.0–34.0)	50(36.0–51.0)
	B (SE)	2.929 (0.983)	0.894 (0.351)	0.822 (0.944)
	p	0.003**	0.011*	0.385
Income (baht/month)	≤4,000	17(11.0–23.8)	30(25.0–32.0)	45(24.0–49.8)
	4,001–8,000	16(8.0–25.0)	30(27.0–32.0)	33(21.0–49.0)
	≥8,001	20(8.5–26.5)	31(29.5–34.5)	41(23.0–49.5)
	B (SE)	0.030 (1.053)	1.369 (0.376)	-0.950 (1.011)
	p	0.977	<0.001**	0.349
Duration in agricultural (years)	≤10	18(11.0–26.0)	30(26.0–33.0)	36(24.0–49.8)
	11–20	15(8.0–22.3)	30(26.0–32.0)	42(21.0–49.0)
	≥21	18(11.0–25.0)	30(27.0–32.0)	46(24.0–51.0)
	B (SE)	-0.997 (1.094)	0.077 (0.390)	1.114 (1.050)
	p	0.363	0.844	0.29
Intercept	B (SE)	14.322 (5.994)	22.018 (2.140)	32.315 (5.756)
	p	0.018*	<0.001**	<0.001**
Model summary	*p*	0.012*	<0.001**	0.093
	Adjusted R^2	0.34	0.66	0.16

Abbreviations: B, the unstandardized regression coefficient; IQR, interquartile range; SE, standard error; significantly different at * $p < 0.05$, **p< 0.01.

Table 3
Comparison of knowledge, attitude, and safety behavior scores by pesticide training participation

Pesticide trained	Frequency(n)	Median (IQR)		
		Knowledge	Attitude	Safety behavior
Never received any training	217	16.0(8.0–24.0)	30.0(26.0–32.0)	28.0(22.0–49.0)
Received training	63	22.0(14.0–27.0)	30.0(28.0–34.0)	49.0(28.0–34.0)
p		0.000**	0.158	0.000**

Abbreviations: IQR, interquartile range; significantly different at $^{**}p < 0.001$ by Mann–Whitney U test.

Table 4
Comparison of knowledge, attitude, and safety behavior scores by level of pesticide risk by reactive paper test

Level of pesticide risk	Frequency (n)	Median (IQR)		
		Knowledge	Attitude	Safety behavior
Normal	78	21.0(12.8–25.3)	30.0(26.0–32.3)	49.0(38.8–50.0)
Safe	160	17.0(9.0–25.0)	30.0(26.0–32.0)	31.0(23.0–49.0)
Risky	42	10.0(6.0–23.0)	30.0(26.8–31.3)	33.5(23.0–48.0)
p		0.009*	0.830	0.002*

Abbreviations: IQR, interquartile range; $^{*}p < 0.05$, Kruskal–Wallis test.

scores, the overall model was significant ($p = 0.012$), explaining 5.2% of the variance (adjusted $R^2 = 0.034$). Education level was a significant predictor ($B = 2.929$, $p = 0.003$), indicating that higher education levels are associated with higher knowledge scores. For attitude scores, the overall model was highly significant ($p < 0.001$), explaining 8.2% of the variance (adjusted $R^2 = 0.066$). Significant predictors included income ($B = 1.369$, $p < 0.001$) and education level ($B = 0.894$, $p = 0.011$), suggesting that higher income and education levels are associated with more positive attitudes. For safety behavior scores, the overall model was not significant ($p = 0.093$), explaining 3.4% of the variance (adjusted $R^2 = 0.016$). Age range was the only significant predictor ($B = 1.944$, $p = 0.041$), indicating that older farmers tend to have better safety behaviors.

The effects of pesticide training on farmers' knowledge, attitudes, and safety behaviors are shown in Table 3. Out of the total 280 participants, a group of 217 farmers who had not undergone any training had median scores of 16.0 (IQR: 8.0–24.0) for knowledge, 30.0 [interquartile range (IQR): 26.0–32.0] for attitude, and 28.0 (IQR: 22.0–49.0) for safety behavior. In contrast, 63 farmers who received training achieved notably higher scores. Specifically, their median scores were 22.0 (IQR: 14.0–27.0) for knowledge, 30.0 (IQR: 28.0–34.0) for attitude, and a particularly elevated score of 49.0 (IQR: 28.0–34.0) for safety behavior. Statistical analyses revealed a significant disparity in the knowledge ($p = 0.000$) and safety behavior ($p = 0.000$) scores between the trained and untrained groups. However, no significant differences were observed in the attitudes ($p = 0.158$). These findings highlight the significant impact of training in improving farmers' knowledge and promoting safety practices. However, there was no significant effect on the attitudes toward pesticide use.

This study analyzed the knowledge, attitude, and safety behavior scores of farmers according to their level of pesticide risk exposure as presented in Table 4. It was found that farmers who were exposed to a normal level of pesticide risk had a significantly higher median knowledge score (21.0, IQR: 12.8–25.3) than farmers who were exposed to safe (17.0, IQR: 9.0–25.0) or risky (10.0, IQR: 6.0–23.0) levels ($p = 0.009$). All groups exhibited a similar median score of approximately 30.0, with an IQR of 26.0–32.3 in terms of attitude, with no notable variations observed among the different risk levels ($p = 0.830$). The safety behavior scores showed significant variations based on the risk level. Farmers at normal risk had the highest median safety behavior score (49.0, IQR: 38.8–50.0), whereas those at safe and risky levels had lower scores of 31.0 (IQR: 23.0–49.0) and 33.5 (IQR: 23.0–48.0), respectively) ($p = 0.002$). These findings indicate a significant difference in knowledge and safety behavior among farmers related to the level of pesticide risk, whereas attitudes toward pesticide use remained consistent across the different risk categories.

To better understand the relationship between these variables, we examined the correlation coefficients between knowledge, attitude, and safety behavior. Finding these correlations helps determine how improvements in knowledge and attitudes might influence safety behavior, aiding in the design of effective interventions and educational programs. Previous research indicates that knowledge often has a more direct impact on behavior. Therefore, controlling one variable allows us to isolate and understand the unique contribution of the other in promoting safe practices. The correlation coefficients between knowledge and attitude and the safety behavior of farmers regarding pesticide use are presented in Table 5. This study had several significant findings. The median knowledge score was 17.0, with an IQR of 9.0–25.0. This knowledge score had a strong positive correlation with safety behavior ($r = 0.493$, $p < 0.001$), indicating that farmers with higher knowledge levels were more likely to engage in safe behavior practices. When controlling for attitude, the correlation between knowledge and safety behavior remained significant ($r = 0.349$, $p < 0.001$), although it was slightly reduced, suggesting that knowledge is a robust predictor of safety behavior even when considering the influence of attitude.

Table 5
Correlation between knowledge, attitude, and safety behavior score

Variable	Median (IQR)	Correlation coefficient					
		Knowledge		Attitude		Safety behavior	
		Not controlled	Controlled	Not controlled	Controlled	Not controlled	Controlled
Knowledge	17.0(9.0–25.0)	—	—	0.196*	0.063[*]	0.493**	0.349**[†]
Attitude	30.0(26.0–32.0)	0.196*	0.063[*]	—	—	0.143*	0.069[‡]
Safety behavior	42.0(23.0–49.0)	0.493**	0.349**[†]	0.143*	0.069[‡]	—	—

Abbreviations: IQR, interquartile range; significantly different at $^{*}p < 0.05$, $^{**}p < 0.001$ by Spearman's rho and partial correlation.
[*] controlled for behavior.
[†] controlled for attitude.
[‡] controlled for knowledge.

The median attitude score was 30.0 (IQR: 26.0–32.0), which showed a weaker positive correlation with safety behavior ($r = 0.143$, $p = 0.017$). This relationship suggests that, while attitude is associated with safety behavior, it is not as strong a predictor as knowledge. Moreover, when the effect of knowledge was controlled for, the correlation between attitude and safety behavior became insignificant ($r = 0.069$, $p = 0.251$), indicating that the direct influence of attitude on safety behavior may be less critical than knowledge.

The correlation between knowledge showed a weaker positive correlation with attitude ($r = 0.196$, $p = 0.001$). Moreover, when the effect of safety behavior was controlled, the correlation between knowledge and attitude became insignificant ($r = 0.063$, $p = 0.293$).

Overall, these results underscore the importance of knowledge as a determinant of safety behavior. Enhancing farmers' knowledge through targeted education and training programs can significantly improve safe pesticide handling. These findings suggest that interventions to improve safety behaviors should focus more on increasing knowledge rather than shifting attitudes.

4. Discussion

As a comprehensive examination of farmers' knowledge, attitudes, and practices regarding pesticide use, our study integrates and extends the existing research in several critical areas. The demographic makeup of our participants, primarily middle-aged to older farmers with a primary school level of education, echoes the findings of the previous studies [15,16]. This demographic profile is significant as it influences the comprehension and implementation of pesticide safety practices, a crucial aspect for both human health and environmental protection.

The significant disparities in knowledge and attitudes across educational levels observed in our study are consistent with the findings of a study conducted in the Wei River catchment in China, which also revealed how educational background profoundly influences farmers' understanding and responses to pesticide-related risks [17]. The role of educational interventions in enhancing knowledge and attitudes toward pesticide safety, as highlighted in previous studies, was also evident in our findings. These interventions significantly improved knowledge and attitudes, paralleling the enhancements seen in Thai orange farmers and Vietnamese biogas users [18,19].

This study highlights a complex relationship between income and attitudes toward pesticide use among farmers, where higher income levels affect attitudes, but do not necessarily translate into safer pesticide practices. This discrepancy may be due to cultural beliefs, social norms, or individual experiences. Thus, while income plays a significant role in shaping attitudes toward pesticide safety, it does not solely drive the adoption of safe practices. This underscores the importance of adopting holistic educational and policy interventions to address the varied motivations and constraints experienced by farmers at different income levels.

The study found that farmers exposed to normal levels of pesticides had higher knowledge and demonstrated safer behaviors than those exposed to lower or higher levels. This supports findings from previous research, including a study in Nakhon Nayok, Thailand, which confirmed a strong correlation among knowledge, attitudes toward pesticides, and their actual usage [20]. Furthermore, rice farmers who used pesticides had a significantly higher prevalence of abnormal serum cholinesterase levels than those who followed proper pesticide use. Other studies that have examined the different health effects and risk perceptions associated with specific types of pesticides and varying levels of exposure have emphasized the complex nature of managing pesticide risks [4,21].

The study revealed a significant and positive correlation between farmers' knowledge of pesticides and their safety behavior, indicating that increased understanding leads to safer practices. This finding is consistent with that of the previous studies. However, the relationship between attitudes and safety behaviors is more complex. Initially, a modest positive correlation was observed, but this became insignificant after adjusting for knowledge, suggesting that, while positive attitudes toward pesticide safety are important, knowledge plays a more critical role in determining safe practices. This aligns with observations from a study in Bangladesh, which emphasized that education and awareness have a greater impact on safety behaviors than attitudes alone [22]. This study underscores the essential role that knowledge plays in ensuring that farmers adopt safe pesticide practices. While positive attitudes toward pesticide safety are beneficial, a strong knowledge base is critical for translating these attitudes into safe behavior. This finding is crucial for developing interventions and educational programs aimed at reducing pesticide-related hazards, emphasizing the need to enhance farmers' understanding of pesticides, and foster favorable safety attitudes. Additionally, a previous quasi-experimental study highlighted the effectiveness of a participatory program in improving pesticide use behavior and reducing blood AChE levels among participants. This program not only increased knowledge and reduced pesticide contamination but also raised awareness of chemical toxicity and changed behavior toward chemicals. Hence, participatory processes are effective in addressing health-related chemical hazards [23]. Similar to this study, among female farmers in Nan, Thailand, the aim was to reduce pesticide exposure among farming mothers. The May–October 2020 study included 78 mothers from two subdistricts. Pesticide behavioral change training was administered to an intervention group of mothers and a control group. Questionnaires and acetylcholinesterase and butyrylcholinesterase levels were used to gauge the effectiveness of training. Pesticide safety knowledge, attitudes, and practices improved and butyrylcholinesterase levels increased, indicating reduced pesticide exposure. The study found that the training program worked and could be used in other agricultural sectors [24].

The study's limitations include the use of purposive sampling, which may limit generalizability to other regions. The cross-sectional design restricts causal inference, requiring longitudinal studies for confirmation. Additionally, the focus on the San Pa Tong district of Chiang Mai may not reflect other agricultural regions. While the findings provide valuable insights, further studies in different areas and diverse farming communities are necessary for broader generalization.

In summary, our study highlighted the critical role of education and training in improving pesticide safety knowledge and practices. They also highlight the importance of considering demographic factors such as age, years of farming experience, and income level when planning and implementing interventions aimed at promoting safer pesticide use. Additionally, encouraging interactions among people and improving overall educational opportunities could be effective strategies in promoting agricultural safety. Future research should focus on the design and testing of such interventions and explore other potential factors affecting safety behaviors in this population. Specifically, these studies could develop targeted educational and training programs to enhance knowledge and attitudes toward pesticide use. Additionally, it is important to explore other potential factors that may influence safety behaviors, such as cultural beliefs, access to safety equipment, social norms, and economic constraints. By addressing these diverse factors, future research can contribute to more comprehensive and effective strategies for promoting safer agricultural practices.

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Data statement

Owing to the confidential questions posed in the survey, participants were guaranteed that the raw data would remain private and would not be disclosed.

CRediT authorship contribution statement

Anurak Wongta: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Nootchakarn Sawarng:** Writing – review & editing, Validation, Methodology, Formal analysis. **Phannika Tongchai:** Validation. **Pichamon Yana:** Data curation. **Surat Hongsibsong:** Writing – review & editing, Visualization, Supervision, Resources, Conceptualization.

Conflicts of interest

The authors declare no conflicts of interest.

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